In [1]:
```
import pandas as pd
```

In [2]:
```
df = pd.read_csv("drug200.csv")
print(df.head())
```

```
   Age Sex      BP Cholesterol  Na_to_K   Drug
0   23   F    HIGH        HIGH   25.355  drugY
1   47   M     LOW        HIGH   13.093  drugC
2   47   M     LOW        HIGH   10.114  drugC
3   28   F  NORMAL        HIGH    7.798  drugX
4   61   F     LOW        HIGH   18.043  drugY
```

In [3]:
```
df.shape
```

Out[3]:
```
(200, 6)
```

In [4]:
```
df.dtypes
```

Out[4]:
```
Age              int64
Sex             object
BP              object
Cholesterol     object
Na_to_K        float64
Drug            object
dtype: object
```

In [6]:
```
df.isnull().sum()
```

Out[6]:
```
Age            0
Sex            0
BP             0
Cholesterol    0
Na_to_K        0
Drug           0
dtype: int64
```

In [7]:
```
df.duplicated().sum()
```

Out[7]:
```
np.int64(0)
```

In [8]:
```
df.describe()
```

Out[8]:

	Age	Na_to_K
count	200.000000	200.000000
mean	44.315000	16.084485
std	16.544315	7.223956
min	15.000000	6.269000
25%	31.000000	10.445500
50%	45.000000	13.936500
75%	58.000000	19.380000
max	74.000000	38.247000

In [9]:
```
df["Sex"].value_counts()
```

Out[9]:
```
Sex
M    104
F     96
Name: count, dtype: int64
```

In [10]:
```
df["BP"].value_counts()
```

Out[10]:
```
BP
HIGH      77
LOW       64
NORMAL    59
Name: count, dtype: int64
```

In [11]:
```
df["Cholesterol"].value_counts()
```

Out[11]:
```
Cholesterol
HIGH      103
NORMAL     97
Name: count, dtype: int64
```

In [12]:
```
df["Drug"].value_counts()
```

Out[12]:
```
Drug
drugY    91
drugX    54
drugA    23
drugC    16
drugB    16
Name: count, dtype: int64
```

In [13]:
```
df.columns
```

Out[13]:
```
Index(['Age', 'Sex', 'BP', 'Cholesterol', 'Na_to_K', 'Drug'], dtype='object')
```

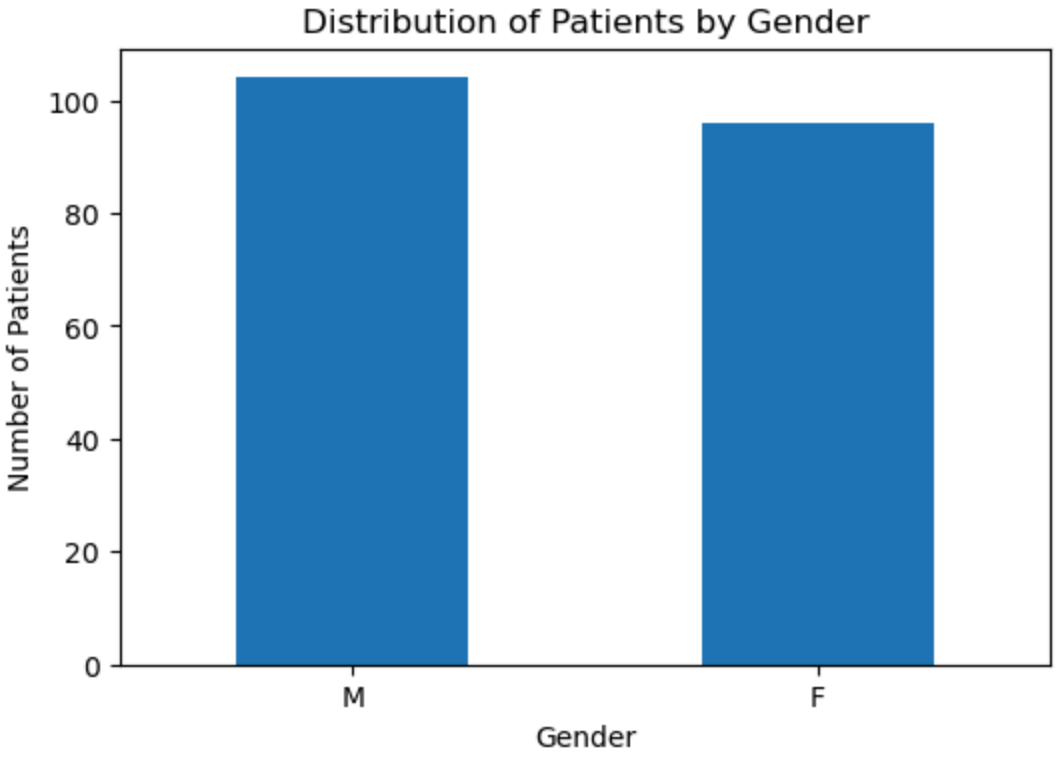
In [14]:
```
df['Age'].mean()
```

Out[14]: np.float64(44.315)

In [16]:
```
gender_count = df["Sex"].value_counts()
```

In [17]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [18]:
```
plt.figure(figsize=(6, 4))
gender_count.plot(kind="bar")
plt.title("Distribution of Patients by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Patients")
plt.xticks(rotation=0)
plt.show()
```



In [19]:
```
drug_count = df["Drug"].value_counts()

print("\nNumber of prescriptions for each drug:")
print(drug_count)

print("\nDrug with highest number of prescriptions:")
print(drug_count.idxmax(), "=", drug_count.max())
```

```
Number of prescriptions for each drug:
Drug
drugY    91
drugX    54
drugA    23
drugC    16
drugB    16
Name: count, dtype: int64

Drug with highest number of prescriptions:
drugY = 91
```

In [20]:

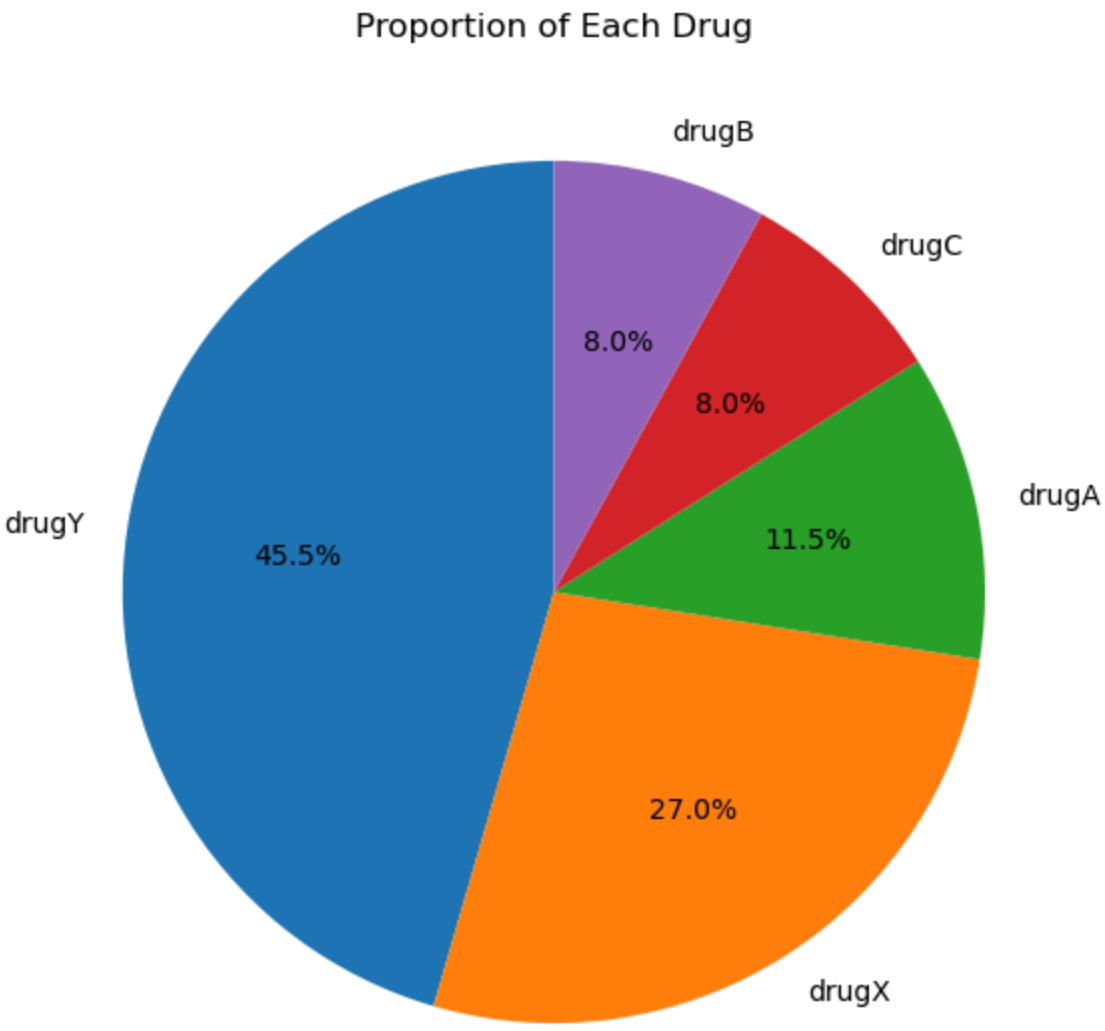
```python
drug_proportion = df["Drug"].value_counts(normalize=True) * 100

print("\nProportion of each drug:")
print(drug_proportion)

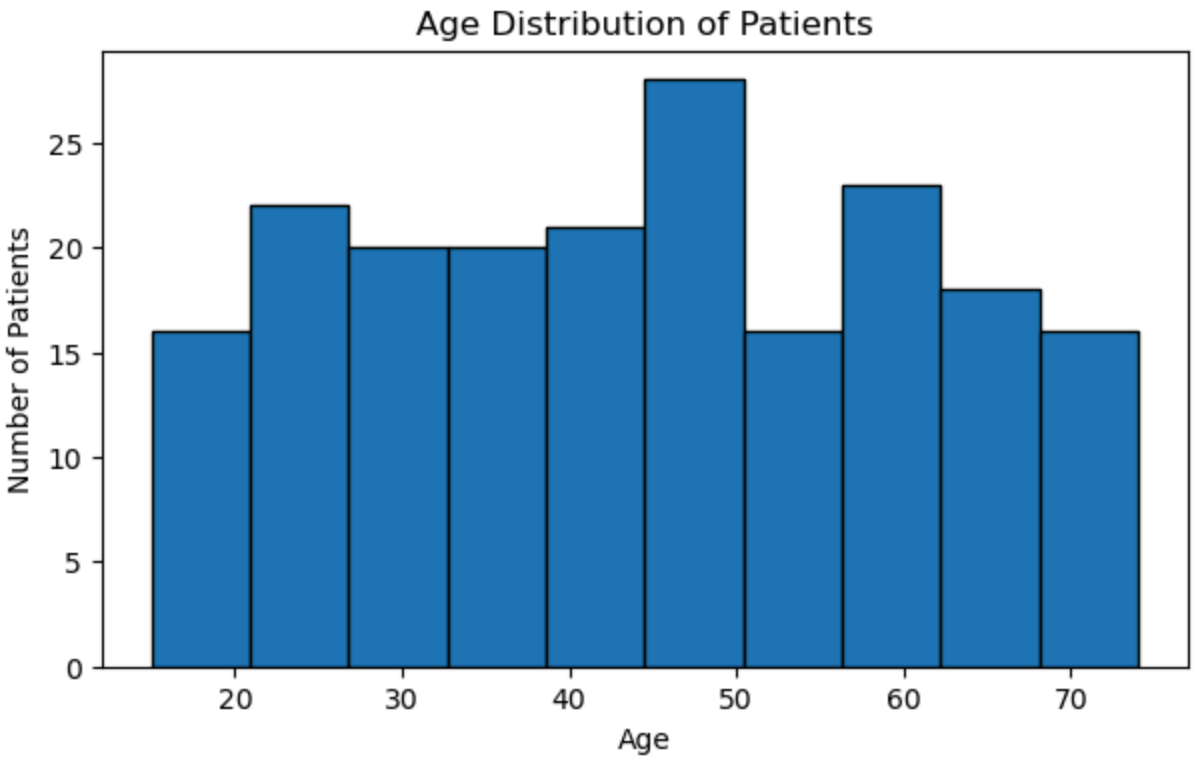
plt.figure(figsize=(7, 7))
plt.pie(
    drug_proportion,
    labels=drug_proportion.index,
    autopct="%1.1f%%",
    startangle=90
)

plt.title("Proportion of Each Drug")
plt.show()
```

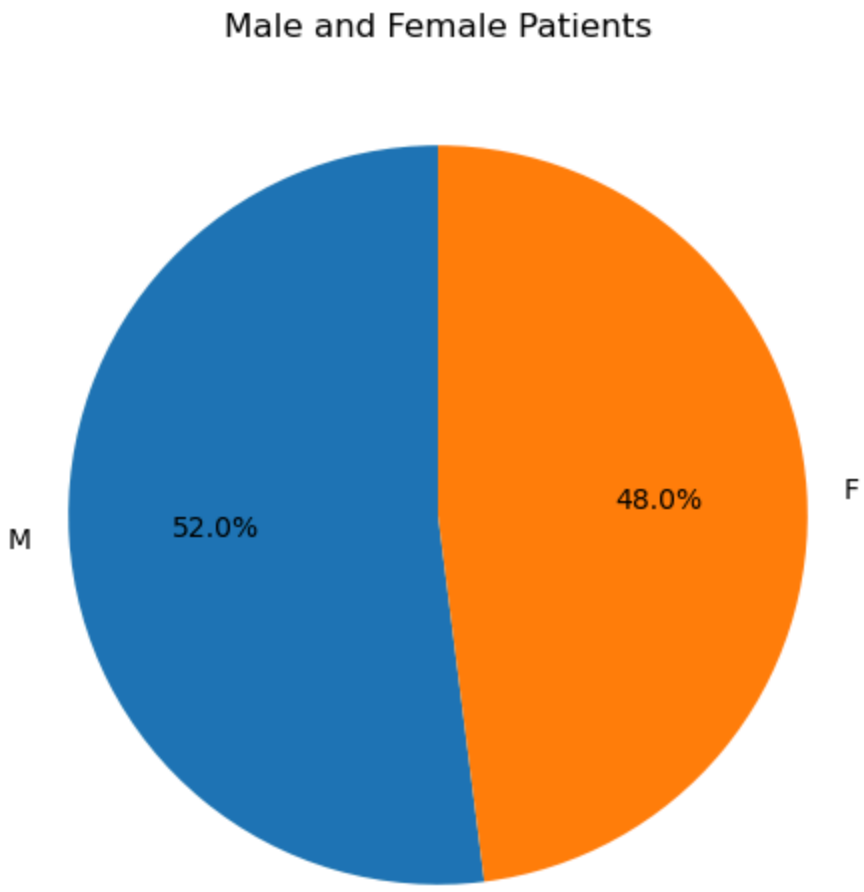
```
Proportion of each drug:
Drug
drugY    45.5
drugX    27.0
drugA    11.5
drugC     8.0
drugB     8.0
Name: proportion, dtype: float64
```



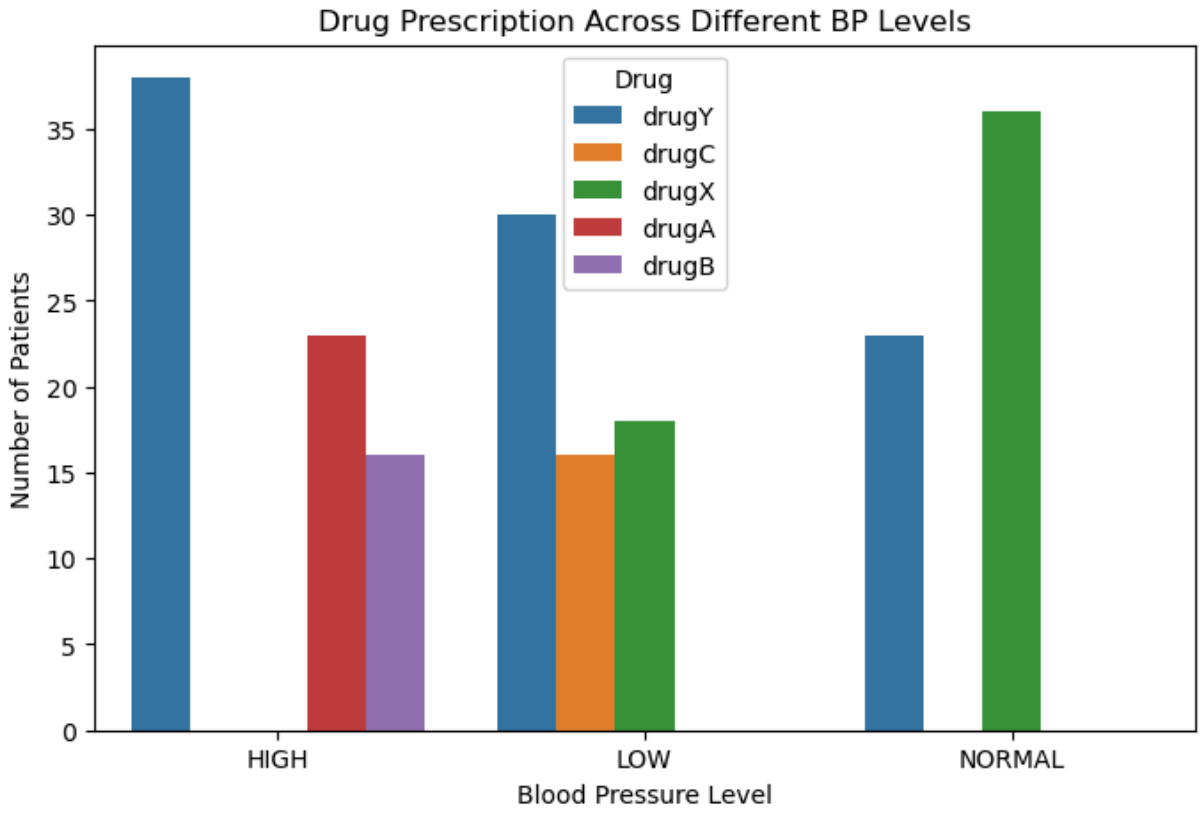
In [21]:
```
plt.figure(figsize=(7, 4))
plt.hist(df["Age"], bins=10, edgecolor="black")
plt.title("Age Distribution of Patients")
plt.xlabel("Age")
plt.ylabel("Number of Patients")
plt.show()
```



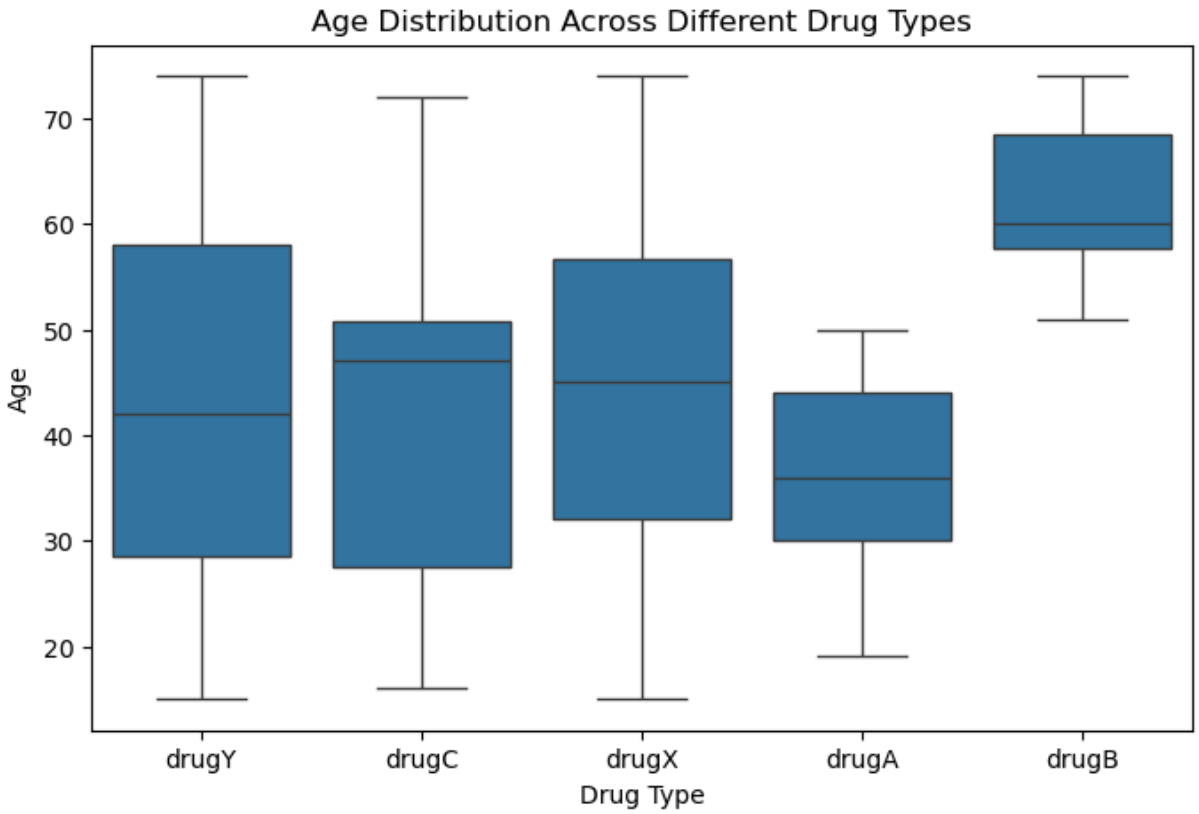
In [22]:
```
plt.figure(figsize=(6, 6))
plt.pie(
    gender_count,
    labels=gender_count.index,
    autopct="%1.1f%%",
    startangle=90
)
plt.title("Male and Female Patients")
plt.show()
```



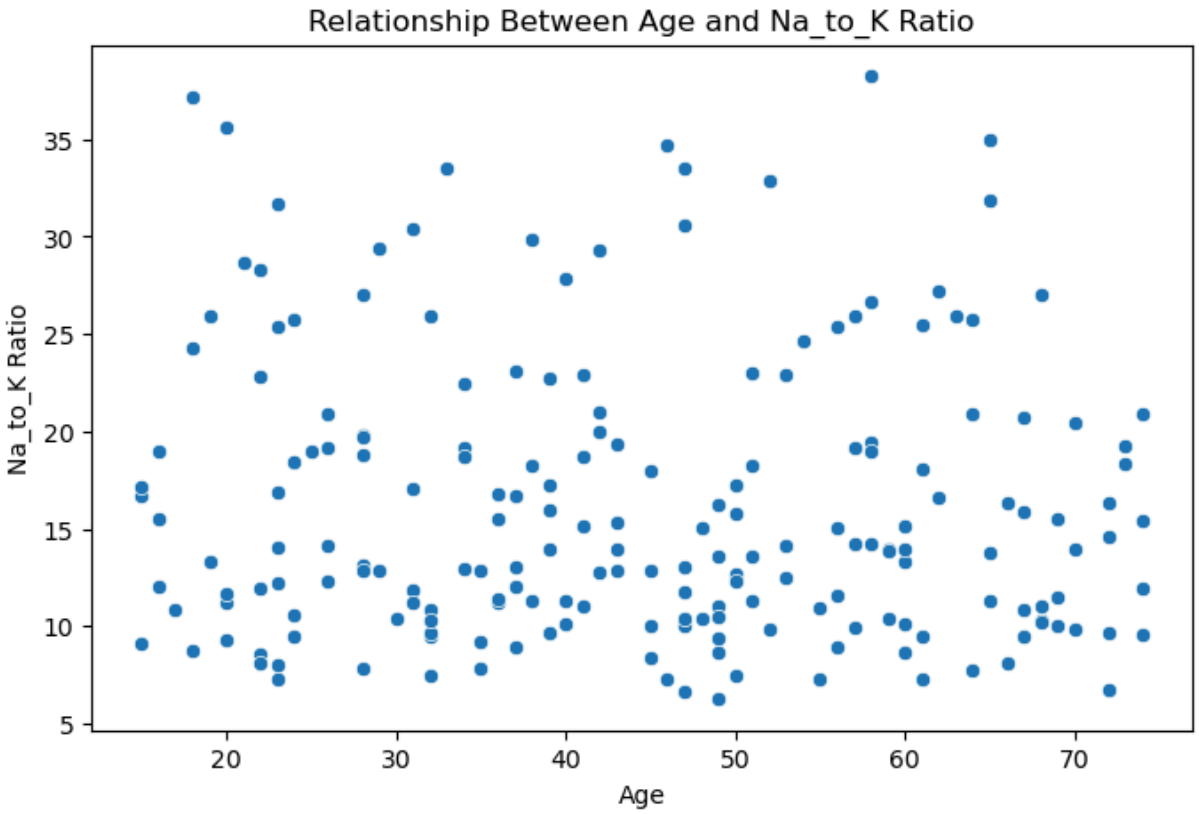
In [23]:
```
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x="BP", hue="Drug")
plt.title("Drug Prescription Across Different BP Levels")
plt.xlabel("Blood Pressure Level")
plt.ylabel("Number of Patients")
plt.show()
```



In [24]:
```
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x="Drug", y="Age")
plt.title("Age Distribution Across Different Drug Types")
plt.xlabel("Drug Type")
plt.ylabel("Age")
plt.show()
```



In [25]:
```
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x="Age", y="Na_to_K")
plt.title("Relationship Between Age and Na_to_K Ratio")
plt.xlabel("Age")
plt.ylabel("Na_to_K Ratio")
plt.show()
```


Relationship Between Age and Na_to_K Ratio
Na_to_K Ratio
Age
5
10
15
20
25
30
35
20
30
40
50
60
70

In [ ]: